

NT-C-019 | Web Development / Web Designing | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Web & Creative Technology
Target learners	Learners seeking front-end web development or web-design roles.
Prerequisites	Basic computer and internet use.
Approved duration	12-16 weeks / 140-180 guided hours.
Final outcome	Design, build, test and deploy responsive, accessible websites with interactive features.
Assessment profile	Creative

Approved tools / platforms

HTML, CSS, JavaScript, responsive design, accessibility tools, Git, browser developer tools; optional design/prototyping tool.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require the brief, working/source files, design rationale, iteration evidence, technical quality and accessible export.	Do not treat a copied template or final export alone as proof of original process and ownership.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-019 | Web Development / Web Designing | NT-C-019-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Internet, HTML semantics and page structure; Selectors, box model, typography and colour
Completion evidence	Semantic pages; Styled pages
Section weight	20% of the weighted mock-test average
Assessment	8 knowledge items (20 marks) + 2 critique/scenario items (20) + 1 timed artifact brief with rubric (60)

Module 1 - Web foundation

Objective: Internet, HTML semantics and page structure

Observable evidence: Semantic pages

Concept	Approved knowledge statement	Application behaviour	Evidence
Internet	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when internet is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Semantic pages
HTML semantics	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when html semantics is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Semantic pages
page structure	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when page structure is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Semantic pages

Module 2 - CSS design

Objective: Selectors, box model, typography and colour

Observable evidence: Styled pages

Concept	Approved knowledge statement	Application behaviour	Evidence
Selectors	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when selectors is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Styled pages
box model	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when box model is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Styled pages
typography	The structured use of type to support hierarchy, readability and visual meaning. In Web Development / Web Designing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when typography is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Styled pages
colour	The controlled use of hue, value and contrast for communication, consistency and accessibility. In Web Development / Web Designing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when colour is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Styled pages

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Semantic pages; Styled pages, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-019 | Web Development / Web Designing | NT-C-019-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Flexbox, Grid and mobile-first design; Syntax, DOM, events and validation
Completion evidence	Responsive site; Interactive page
Section weight	25% of the weighted mock-test average
Assessment	8 knowledge items (20 marks) + 2 critique/scenario items (20) + 1 timed artifact brief with rubric (60)

Module 3 - Responsive layout

Objective: Flexbox, Grid and mobile-first design

Observable evidence: Responsive site

Concept	Approved knowledge statement	Application behaviour	Evidence
Flexbox	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when flexbox is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Responsive site
Grid	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when grid is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Responsive site
mobile-first design	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when mobile-first design is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Responsive site

Module 4 - JavaScript

Objective: Syntax, DOM, events and validation

Observable evidence: Interactive page

Concept	Approved knowledge statement	Application behaviour	Evidence
Syntax	The formal rules used to write valid instructions in a programming language. In Web Development / Web Designing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when syntax is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Interactive page
DOM	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dom is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Interactive page
events	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when events is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Interactive page
validation	Checking input or output against defined rules before trusting it. In Web Development / Web Designing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when validation is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Interactive page

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Responsive site; Interactive page, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-019 | Web Development / Web Designing | NT-C-019-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Wireframes, hierarchy, components and assets; Fetch, JSON, forms and error states
Completion evidence	Design specification; Data-driven page
Section weight	25% of the weighted mock-test average
Assessment	8 knowledge items (20 marks) + 2 critique/scenario items (20) + 1 timed artifact brief with rubric (60)

Module 5 - Design workflow

Objective: Wireframes, hierarchy, components and assets

Observable evidence: Design specification

Concept	Approved knowledge statement	Application behaviour	Evidence
Wireframes	A low-fidelity representation of structure, content and interaction. In Web Development / Web Designing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when wireframes is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Design specification
hierarchy	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when hierarchy is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Design specification
components	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when components is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Design specification
assets	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when assets is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Design specification

Module 6 - APIs and forms

Objective: Fetch, JSON, forms and error states

Observable evidence: Data-driven page

Concept	Approved knowledge statement	Application behaviour	Evidence
Fetch	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when fetch is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Data-driven page
JSON	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when json is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Data-driven page
forms	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when forms is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Data-driven page
error states	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when error states is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Data-driven page

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Design specification; Data-driven page, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-019 | Web Development / Web Designing | NT-C-019-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Accessibility, performance, SEO basics and testing; Git, hosting, analytics basics and documentation
Completion evidence	Audit report; Live portfolio website
Section weight	30% of the weighted mock-test average
Assessment	8 knowledge items (20 marks) + 2 critique/scenario items (20) + 1 timed artifact brief with rubric (60)

Module 7 - Quality

Objective: Accessibility, performance, SEO basics and testing

Observable evidence: Audit report

Concept	Approved knowledge statement	Application behaviour	Evidence
Accessibility	Design and implementation that allows people with diverse abilities to use a product. In Web Development / Web Designing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when accessibility is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Audit report
performance	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when performance is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Audit report
SEO basics	Improving discoverability and usefulness for relevant search intent while following search-engine guidance. In Web Development / Web Designing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when seo basics is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Audit report
testing	Systematic verification that expected behaviour and important failure cases work correctly. In Web Development / Web Designing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when testing is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Audit report

Module 8 - Deployment capstone

Objective: Git, hosting, analytics basics and documentation

Observable evidence: Live portfolio website

Concept	Approved knowledge statement	Application behaviour	Evidence
Git	Distributed version control for tracking and reviewing source changes. In Web Development / Web Designing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when git is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Live portfolio website
hosting	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when hosting is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Live portfolio website
analytics basics	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when analytics basics is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Live portfolio website
documentation	A defined competency within Web Development / Web Designing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when documentation is appropriate; apply it in a Web Development / Web Designing task; verify the result.	Live portfolio website

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Audit report; Live portfolio website, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-019 | Web Development / Web Designing | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-019-SK01	Web foundation	Prerequisite foundation	Semantic pages	Critical
NT-C-019-SK02	CSS design	M01 plus earlier foundations	Styled pages	Critical
NT-C-019-SK03	Responsive layout	M02 plus earlier foundations	Responsive site	Supporting
NT-C-019-SK04	JavaScript	M03 plus earlier foundations	Interactive page	Supporting
NT-C-019-SK05	Design workflow	M04 plus earlier foundations	Design specification	Supporting
NT-C-019-SK06	APIs and forms	M05 plus earlier foundations	Data-driven page	Critical
NT-C-019-SK07	Quality	M06 plus earlier foundations	Audit report	Supporting
NT-C-019-SK08	Deployment capstone	M07 plus earlier foundations	Live portfolio website	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-019 | Web Development / Web Designing | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-019-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Web foundation and CSS design?	The learner should provide Semantic pages; Styled pages and explain how the work applies Internet, HTML semantics and page structure; Selectors, box model, typography and colour. Recall alone is insufficient.
NT-C-019-S01-EX02	Scenario/remediation	A learner reports completing CSS design but cannot produce Styled pages. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-019-S01, request Styled pages, and allow a new equivalent test only after remediation.
NT-C-019-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Responsive layout and JavaScript?	The learner should provide Responsive site; Interactive page and explain how the work applies Flexbox, Grid and mobile-first design; Syntax, DOM, events and validation. Recall alone is insufficient.
NT-C-019-S02-EX02	Scenario/remediation	A learner reports completing JavaScript but cannot produce Interactive page. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-019-S02, request Interactive page, and allow a new equivalent test only after remediation.
NT-C-019-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Design workflow and APIs and forms?	The learner should provide Design specification; Data-driven page and explain how the work applies Wireframes, hierarchy, components and assets; Fetch, JSON, forms and error states. Recall alone is insufficient.
NT-C-019-S03-EX02	Scenario/remediation	A learner reports completing APIs and forms but cannot produce Data-driven page. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-019-S03, request Data-driven page, and allow a new equivalent test only after remediation.
NT-C-019-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Quality and Deployment capstone?	The learner should provide Audit report; Live portfolio website and explain how the work applies Accessibility, performance, SEO basics and testing; Git, hosting, analytics basics and documentation. Recall alone is insufficient.
NT-C-019-S04-EX02	Scenario/remediation	A learner reports completing Deployment capstone but cannot produce Live portfolio website. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-019-S04, request Live portfolio website, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Creative blueprint. Human-review every scored item before initial production release.

NT-C-019 | Web Development / Web Designing | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Responsive landing page	HTML, CSS and accessibility	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Interactive directory	JavaScript, APIs and forms	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Business website	Design, testing, SEO and deployment	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require the brief, working/source files, design rationale, iteration evidence, technical quality and accessible export.

NT-C-019 | Web Development / Web Designing | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Junior Web Developer	HTML 25% [critical]; CSS 25% [critical]; JavaScript 25%; Git 25%	Live website	If a critical skill is below 60 or live website is missing, return Improve First or Needs Evidence with the linked module/project.
2	Front-end Developer Trainee	Responsive UI 34% [critical]; DOM 33% [critical]; APIs 33%	Interactive project	If a critical skill is below 60 or interactive project is missing, return Improve First or Needs Evidence with the linked module/project.
3	Web Designer	Layout 34% [critical]; typography 33% [critical]; responsive design 33%	Design case study	If a critical skill is below 60 or design case study is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-019 | Web Development / Web Designing | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Web Development / Web Designing for?	Learners seeking front-end web development or web-design roles.
What should I know before starting?	Basic computer and internet use.
How long is the approved learning journey?	12-16 weeks / 140-180 guided hours.
What will I be able to demonstrate?	Design, build, test and deploy responsive, accessible websites with interactive features.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Web Development / Web Designing knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

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course_code=NT-C-019; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```